JavaScript Revision Lab - Arrays, Strings, Functions

Lab 1: Student Marks Analyzer (1D Array + Functions)

Create a function analyzeMarks(marks) that accepts an array of student marks.
Tasks:
1. Find the highest mark.
2. Find the lowest mark.
3. Calculate the average mark.
4. Return all results from the function.

Example Input:
[78, 92, 65, 88, 95, 70]

Expected Output:
Highest: 95
Lowest: 65
Average: 81.33

Lab 2: String Utility Toolkit (String + Functions)

Create functions to perform the following operations on a given string:
1. Count total characters.
2. Count vowels.
3. Reverse the string.
4. Check whether the string is a palindrome.

Example Input:
'madam'

Expected Output:
Characters: 5
Vowels: 2
Reverse: madam
Palindrome: Yes

Lab 3: Product Sales Report (2D Array + Functions)

A 2D array contains sales of products across 4 weeks.

Example:
[
 [100, 120, 130, 150],
 [80, 90, 100, 110],
 [200, 180, 220, 250]

]

Create functions to:
1. Calculate total sales for each product.
2. Find the product with highest total sales.
3. Calculate overall sales.

Lab 4: Sentence Analyzer (Array + String + Functions)

Accept a sentence from the user.

Tasks:
1. Convert the sentence into an array of words.
2. Count total words.
3. Find the longest word.
4. Count occurrences of a given word.
5. Display all results using functions.

Example:
'JavaScript functions are powerful and reusable'

Lab 5: Movie Seat Booking System (2D Array + String + Functions)

Create a cinema seating system using a 2D array.

Example:
```
[
 ['A1','A2','A3'],
 ['B1','B2','B3'],
 ['C1','C2','C3']
]
```

Tasks:
1. Create a function to book a seat.
2. Replace booked seat value with 'BOOKED'.
3. Display available seats.
4. Count total available seats.
5. Search whether a seat exists or not.